SELECTION PROJECT BRIEF

Food & Hospitality KPI Intelligence Dashboard

Live Working Prototype Development Challenge

Document Type	Domain	Tier
Selection Project Brief	Food & Hospitality Tech	Frontend-First / Full-Stack Optional

1. Project Overview

The hospitality industry runs on instinct far more than it should. Owners and managers who cannot see their business in real time — who rely on end-of-week reports and gut feel — consistently leave 10–20% of revenue uncaptured, overspend on labour, and discover problems after they have already become expensive habits.

This project challenges you to build a live, interactive KPI Intelligence Dashboard for food and hospitality businesses — a tool that functions as the operational nerve centre for a restaurant, café, hotel F&B outlet, or cloud kitchen. The dashboard must surface day-to-day operational data alongside strategic intelligence and smart alerts that prompt action before a problem becomes costly.

The objective is to demonstrate your ability to think like a product builder: to take domain knowledge, translate it into meaningful data experiences, and deliver a prototype that a real business owner could sit in front of and immediately understand the story of their business today.

1.1 The Problem Being Solved

Most hospitality businesses today operate using one of three inadequate approaches:

- Paper or whiteboard tallies that capture revenue but miss the intelligence layer entirely
- Generic POS reports that are cluttered, backward-looking, and require interpretation
- Spreadsheet dashboards that are fragile, manually maintained, and inaccessible to the team on the floor

None of these answers the questions that matter in real time: Are we on track today? Where is the bottleneck right now? Should I stop accepting delivery orders for the next 30 minutes? What is my most profitable item, and is it being sold to its potential?

1.2 The Solution Vision

A single-screen, role-aware dashboard that presents 49 categorised KPIs across 7 business domains. The interface must be immediately readable by a non-technical restaurant manager, while offering enough depth for an owner or investor to make strategic decisions. It will include three tiers of KPI intelligence:

Tier	Description	Example
Operational	**Day-to-day metrics reviewed every shift**	**Kitchen ticket time, covers served, food cost %**
Strategic	**Monthly growth and revenue levers**	**Customer lifetime value, staff turnover, prime cost %**

Tier	Description	Example
Smart Alerts	**Automated threshold triggers requiring action**	**Delivery time >40min, cancellation rate spike >5%**

2. KPI Knowledge Bank

The following table is the complete reference for all 49 KPIs your dashboard must support. Use this as the single source of truth for data modelling, mock data generation, and display logic. Each KPI includes its formula, classification tier, alert threshold where applicable, and industry benchmark target.

Domain	KPI Name	Formula / Method	Tier	Alert Threshold	Benchmark
Revenue	**Daily Revenue Target Attainment**	*Actual ÷ Target × 100*	**Operational**	—	>90%
Revenue	**Revenue Per Available Seat Hour (RevPASH)**	*Total Revenue ÷ (Seats × Op. Hours)*	**Smart Alert**	Drop >15% vs same weekday LW	$15–$25/seat-hr
Revenue	**Average Check / Spend Per Cover**	*Total Revenue ÷ Total Covers*	**Strategic**	—	Grows 5–8% YoY
Revenue	**Revenue by Channel Mix**	*Channel Rev ÷ Total Rev × 100*	**Operational**	—	Track split by dine-in/delivery/takeaway
Revenue	**Peak Hour Revenue Capture Rate**	*Peak 2hr Rev ÷ Daily Rev*	**Smart Alert**	Below 35% of daily revenue	>35% in peak window
Revenue	**Catering & Events Revenue %**	*Events Rev ÷ Total Rev × 100*	**Strategic**	—	Grows monthly
Operations	**Table Turnover Rate**	*Covers ÷ Number of Seats*	**Operational**	—	Fast casual: 3–5x
Operations	**Average Table Dwell Time**	*Total Seated Min ÷ Covers*	**Smart Alert**	Exceeds target by >20 min at peak	Varies by format
Operations	**Kitchen Ticket Time (KTT)**	*Order placed → Food on pass (avg)*	**Operational**	—	<12 min casual, <22 min fine dining
Operations	**Order to Delivery Time**	*Order confirmed → Delivered (avg min)*	**Smart Alert**	Rolling 30-min avg >40 min	<35 minutes
Operations	**Seat Utilisation Rate**	*Occupied Seat Hrs ÷ Available Seat Hrs*	**Operational**	—	>70% during service
Operations	**Dead Slot Identification Index**	*Revenue per 30-min block / 7-day heatmap*	**Strategic**	—	Informs promo scheduling
Orders & Menu	**Menu Item Contribution Margin**	*(Sell Price − Food Cost) per item ranked*	**Strategic**	—	Stars = high margin + high popularity

Domain	KPI Name	Formula / Method	Tier	Alert Threshold	Benchmark
Orders & Menu	**Order Error Rate**	*Wrong Orders ÷ Total Orders × 100*	**Smart Alert**	>2% in any shift	<1%
Orders & Menu	**Cancellation & Abandoned Order Rate**	*Cancelled ÷ Total Orders × 100*	**Smart Alert**	Spike above 5% in 2-hr window	<3%
Orders & Menu	**Upsell Conversion Rate**	*Orders with Upsell ÷ Total Orders × 100*	**Operational**	—	35–50% (trained staff)
Orders & Menu	**Item 86 Frequency (Stock-Out Rate)**	*Times 86d ÷ Total Service Periods*	**Strategic**	—	<2x/week per item
Orders & Menu	**Beverage-to-Food Revenue Ratio**	*Beverage Rev ÷ Food Rev × 100*	**Operational**	—	30–40% beverage of total
Staff	**Labour Cost % of Revenue**	*Labour Cost ÷ Revenue × 100*	**Operational**	—	28–35%
Staff	**Revenue Per Labour Hour**	*Total Revenue ÷ Total Staff Hours*	**Smart Alert**	Drops >10% vs last month same volume	Track trending
Staff	**Staff Turnover Rate**	*(Staff Left ÷ Avg Headcount) × 100*	**Strategic**	—	<30%/yr
Staff	**Covers Per Server Per Shift**	*Total Covers ÷ FOH Staff on Shift*	**Operational**	—	25–40 covers/server
Staff	**No-Show & Late Rate (Staff)**	*Unplanned Absences ÷ Scheduled Shifts × 100*	**Smart Alert**	Trigger 2hrs before shift start	<3%
Staff	**Training Hours Per Staff Per Month**	*Total Training Hrs ÷ Staff Count*	**Operational**	—	4–8 hrs/month
Customer	**Guest Return Rate**	*Returning Customers ÷ Total Unique Customers × 100*	**Strategic**	—	>40% monthly
Customer	**Net Promoter Score (NPS)**	*% Promoters (9–10) − % Detractors (0–6)*	**Operational**	—	>50 good, >70 excellent
Customer	**Online Review Score & Velocity**	*Avg rating + review frequency*	**Smart Alert**	Any 1–2 star review triggers instant alert	4.4+ across platforms
Customer	**Customer Lifetime Value (CLV)**	*Avg Spend × Visits/Yr × Avg Years*	**Strategic**	—	Informs acquisition spend cap
Customer	**Complaint Resolution Time**	*Complaint raised → Resolved (avg min)*	**Smart Alert**	Unresolved after 20 min in-	<15 min in-house

Domain	KPI Name	Formula / Method	Tier	Alert Threshold	Benchmark
				house / 4hr online	
Customer	**Reservation No-Show Rate**	*No-Show Reservations ÷ Total × 100*	**Operational**	—	<8%
Inventory	**Food Cost Percentage**	*COGS ÷ Food Revenue × 100*	**Operational**	—	28–35% food, 18–24% bev
Inventory	**Food Waste %**	*Waste Value ÷ Total Purchased × 100*	**Smart Alert**	Daily waste value exceeds pre-set threshold	<3% of purchased cost
Inventory	**Inventory Turnover Rate**	*COGS ÷ Average Inventory Value*	**Strategic**	—	4–8x/month perishables
Inventory	**Shrinkage & Variance Rate**	*(Expected − Actual) ÷ Expected × 100*	**Smart Alert**	Weekly variance >2% on high-value category	<1%
Inventory	**Supplier On-Time Delivery Rate**	*On-Time Deliveries ÷ Total Orders × 100*	**Operational**	—	>95% on-time
Inventory	**Cost Per Portion Accuracy**	*Actual plated cost vs recipe costed (variance %)*	**Strategic**	—	<5% variance
ROI & Growth	**Prime Cost (Labour + COGS) %**	*(Labour + COGS) ÷ Revenue × 100*	**Strategic**	—	<60% (target 55%)
ROI & Growth	**EBITDA Margin**	*(Rev − COGS − Labour − OpEx) ÷ Rev × 100*	**Strategic**	—	10–15%
ROI & Growth	**Marketing ROI by Channel**	*(Channel Rev − Cost) ÷ Cost × 100*	**Strategic**	—	Track per channel
ROI & Growth	**Break-Even Covers Per Day**	*Fixed Daily Costs ÷ Avg Contribution Margin*	**Operational**	—	Display as daily team target
ROI & Growth	**Cash Flow Runway (Days)**	*Cash Balance ÷ Avg Daily Outflow*	**Smart Alert**	Runway drops below 21 days	>45 days healthy
ROI & Growth	**Loyalty Programme ROI**	*(Incremental Rev − Programme Cost) ÷ Cost*	**Strategic**	—	Members spend 18–25% more
ROI & Growth	**Delivery Platform Commission Impact**	*Delivery Rev × Commission % = Net Delivery Rev*	**Smart Alert**	Net delivery margin <8%	>12% net margin on delivery

3. Technical Specifications

3.1 Core Deliverable

A live, browser-accessible frontend prototype of the KPI dashboard. The prototype must function as a real working demo — interactive, responsive, and populated with realistic mock data that tells a coherent story of a hospitality business over a day, week, and month.

The prototype does NOT need to connect to a real POS or live data source. All data may be simulated using a local mock data layer (JSON, JS objects, or a lightweight mock API). The quality of the data simulation — whether it feels real — is part of the evaluation.

3.2 Mandatory Frontend Requirements

#	Requirement	Acceptance Criteria
F1	Multi-domain navigation	All 7 KPI domains accessible via tabs, sidebar, or nav — no page reload required
F2	KPI card system	Each KPI displayed as a card with current value, trend indicator (up/down/flat), and benchmark status
F3	Smart Alert panel	Dedicated visible alert feed showing all 18 smart alerts — active or simulated as triggered
F4	Time range selector	Toggle between Today / This Week / This Month — values update across all KPIs
F5	Revenue & trend charts	At minimum: daily revenue trend, channel mix chart, RevPASH by time-of-day heatmap
F6	Menu Engineering quadrant	Scatter/quadrant chart mapping menu items to Stars / Plowhorses / Puzzles / Dogs
F7	Mobile responsiveness	Fully usable on a 375px wide screen (iPhone SE). No horizontal scroll.
F8	Alert simulation toggle	Ability to trigger a demo alert (e.g. delivery time spike) to showcase the alert system live
F9	Role-based view switching	At minimum two views: Manager (operational focus) and Owner (strategic + financial focus)
F10	Dark mode support	Dashboard must be fully readable and functional in both light and dark mode

3.3 Bonus Backend Requirements (Plus Points)

These are not mandatory but will significantly differentiate candidates who implement them.

#	Bonus Feature	Implementation Notes
B1	REST API with mock data endpoints	Node/Express or FastAPI serving /api/kpis/:domain, /api/alerts, /api/trends — consumed by the frontend
B2	Persistent alert state	Alerts can be acknowledged/dismissed and state persists across page reloads (localStorage or DB)
B3	Real-time simulation	WebSocket or polling that updates select KPIs every 30–60 seconds to mimic live data
B4	CSV / PDF export	Export any domain's KPI summary as a downloadable report
B5	Threshold configuration UI	A settings panel where alert thresholds can be customised by the user
B6	Database-backed mock data	SQLite or PostgreSQL seeded with 90 days of realistic data with proper time-series queries

3.4 Technology Stack

There is no mandated stack. Candidates are free to choose what they work best with. However, the following stacks are recommended for this use case:

Layer	Recommended Options	Notes
Frontend Framework	React, Vue 3, or Svelte	Vanilla JS acceptable if implementation quality is high
Charts & Dataviz	Recharts, Chart.js, ApexCharts, D3.js	Must support at least bar, line, scatter, and pie/donut types
Styling	Tailwind CSS, CSS Modules, or Styled Components	No CSS frameworks that override too aggressively (e.g. plain Bootstrap without customisation)
Backend (optional)	Node.js + Express, Python + FastAPI, or Next.js API routes	Keep it thin — this is primarily a frontend challenge
Mock Data	Hardcoded JSON, faker.js, or custom seed scripts	Data must feel realistic — use actual hospitality figures from the KPI table in Section 2
Deployment	Vercel, Netlify, Railway, or Render (free tier)	Must be a live URL — local-only submissions will not be evaluated

4. UI/UX Design Specifications

4.1 Design Philosophy

The dashboard is built for people who are not data analysts. A head chef, a floor manager, or a restaurant owner needs to look at this screen mid-service and understand the story of their business in under 10 seconds. Design accordingly.

- Clarity over density — do not try to show all 49 KPIs simultaneously on one screen
- Action-oriented — every smart alert must tell the user what to do, not just what happened
- Contextual benchmarks — show whether a number is good or bad, not just what it is
- Hierarchy — operational numbers are daily, strategic numbers are monthly; the UI must reflect this
- Colour as signal — use green/amber/red sparingly and consistently for KPI status only

4.2 Required Screen Layouts

Screen / View	Must Include	Design Notes
Command Centre (Home)	4–6 top KPI summary cards, live alert feed, daily revenue vs target, quick domain links	First screen on load. Should feel like a cockpit — essential info only
Revenue Domain	RevPASH trend, channel mix donut, daily vs target bar chart, peak hour heatmap	The primary money screen — owners will live here
Operations Domain	KTT gauge/trend, table turnover tracker, seat utilisation timeline, dead-slot heatmap	7-day heatmap is a key differentiator — build it well
Orders & Menu	Menu engineering quadrant, top/bottom items, order error rate, 86 frequency tracker	Quadrant chart is the focal point of this section
Staff Dashboard	Labour cost % gauge, turnover tracker, covers/server chart, no-show log	Sensitive — handle staff-level data carefully in terms of display
Customer Intelligence	Return rate trend, NPS score display, CLV segment, review score tracker	NPS display must show trend over time, not just current score
Inventory & Waste	Food cost % tracker, waste value chart, shrinkage alert log, supplier scorecard	Waste should be shown in both kg and monetary value

Screen / View	Must Include	Design Notes
ROI & Growth	Prime cost gauge, EBITDA trend, break-even covers tracker, marketing ROI bars	Owner-facing. Dense with numbers but must remain readable
Smart Alerts Panel	Full alert log with timestamp, domain tag, severity, and recommended action	Filterable by domain and severity. Most critical alert shows first.

4.3 KPI Card Component Specification

Every KPI card in the system must follow a consistent structure. This ensures the dashboard feels like a product, not a collection of separate widgets.

Element	Required	Detail
KPI Name	Yes	Short version (max 5 words). Full name available on hover/expand.
Current Value	Yes	Prominent — largest text on the card. Formatted with correct unit (%, £/$, min, x).
Trend Arrow	Yes	Up (green/red depending on KPI direction), Down, or Flat vs previous period
Benchmark Indicator	Yes	Colour-coded dot or badge: On Target (green), Watch (amber), Alert (red)
Period Label	Yes	Today / This Week / This Month — matches the global time selector
Sparkline	Recommended	A 7-point mini trend line showing recent trajectory. Can be skipped for count-type KPIs.
Alert Icon	For Smart Alert KPIs only	Bell icon with red badge when threshold is breached. Click expands recommended action.
Formula Tooltip	Recommended	Hover/tap on KPI name reveals the formula from Section 2. Useful for training.

5. Mock Data Specification

Your mock data must tell a coherent story about a real-world restaurant. Generic random numbers that do not follow the patterns of actual hospitality operations will be penalised in evaluation. Below is a reference scenario and the data ranges you should use to build your seed data.

5.1 Reference Business Profile

Attribute	Value
Business Type	Casual-dining restaurant with delivery channel
Seating Capacity	60 covers across 18 tables
Operating Hours	11:00 am – 11:00 pm (12 hrs), 7 days/week
Average Check	$28 dine-in / $22 delivery
Daily Cover Target	150–180 dine-in covers + 40–60 delivery orders
Monthly Revenue Target	$180,000
Staff Count	14 full-time, 8 part-time
Menu Size	42 items across 6 categories
POS System	Simulated — no real integration required

5.2 Realistic Data Ranges by KPI Domain

Use these ranges when generating your mock data. Values should fluctuate realistically: weekends higher than weekdays, lunch and dinner peaks at 12–2pm and 7–9pm, Monday mornings consistently lower.

KPI	Realistic Range	Peak Scenario	Alert Scenario
Daily Revenue	$4,200 – $7,800	$9,200 (Fri/Sat)	Below $3,800 on a weekday
RevPASH	$14 – $22/seat-hr	$28 on Sat dinner	Below $10 Mon 3–5pm
Table Turnover	2.8 – 3.8x	4.5x Fri dinner service	Below 2x = investigation
Kitchen Ticket Time	9 – 16 min	18 min at peak on weekend	>22 min triggers alert
Food Cost %	29 – 34%	—	>36% triggers review
Labour Cost %	28 – 33%	—	>38% triggers alert
NPS Score	48 – 72	78 after menu relaunch	Below 40 = urgent

KPI	Realistic Range	Peak Scenario	Alert Scenario
Order Error Rate	0.6 – 1.8%	—	>2.5% = shift debrief
Food Waste %	2.1 – 4.8%	—	>5% = same-day action
Delivery Time	28 – 38 min	—	>42 min = pause orders
Prime Cost %	54 – 61%	—	>65% = strategic review

5.3 Suggested Mock Data Architecture

- Seed 90 days of daily summary data (all KPI domains) into a JSON file or database
- Add hourly data for the current day to power the real-time/live view
- Include at least 3 pre-built alert scenarios that can be toggled on/off in the demo
- Ensure your menu data includes all four engineering quadrant types (Stars, Plowhorses, Puzzles, Dogs)
- Staff data should show realistic shift patterns — not all 22 staff working every day

6. Evaluation Criteria

Submissions will be evaluated across five dimensions. Each dimension carries a weighted score. The total is 100 points.

Dimension	Weight	What We Are Looking For
Product Thinking	**30 pts**	**Does the dashboard answer real hospitality questions? Does it feel like it was designed for a restaurant owner, not a developer? Is the information hierarchy correct?**
Frontend Implementation	**25 pts**	**Code quality, component structure, reusability, responsive design, performance. Charts must be accurate and readable. No broken layouts.**
KPI Coverage & Accuracy	**20 pts**	**How many of the 49 KPIs from Section 2 are implemented? Are formulas applied correctly to the mock data? Are benchmarks displayed and contextualised?**
Smart Alert System	**15 pts**	**Is the alert panel functional? Can alerts be triggered, viewed, and actioned? Are recommended actions shown? Dismissed alerts handled?**
Data Realism & Storytelling	**10 pts**	**Does the mock data feel like a real restaurant? Do the numbers follow hospitality patterns (peaks, slow days, seasonal dip)? Does the dashboard tell a coherent story?**

6.1 Bonus Points (Up to +20 pts)

Bonus Feature	Points	Criteria
Working REST API backend	+7 pts	Fully functional, consumed by the frontend, documented
Real-time simulation (WebSocket / polling)	+5 pts	Live KPI values updating during demo without page reload
Threshold configuration UI	+4 pts	User can adjust alert thresholds from within the dashboard
CSV/PDF export	+2 pts	At least one domain exports a meaningful downloadable report
Accessibility (WCAG AA compliance)	+2 pts	Keyboard nav, ARIA labels, colour contrast passing AA

7. Submission Requirements

7.1 What to Submit

1. A live URL (Vercel / Netlify / Railway / Render) where the demo is accessible without login
2. A public GitHub repository containing all source code
3. A README.md covering: setup instructions, tech stack summary, architecture decisions, known limitations, and a list of which KPIs from Section 2 are implemented
4. A 3–5 minute screen recording walking through the dashboard and demonstrating at least 2 smart alert scenarios

7.2 Submission Checklist

	Item	Status
[ ]	Live demo URL is working and accessible	Mandatory
[ ]	GitHub repo is public with clean commit history	Mandatory
[ ]	All 10 mandatory frontend requirements (F1–F10) implemented	Mandatory
[ ]	At minimum 30 of 49 KPIs from Section 2 are visible in the dashboard	Mandatory
[ ]	Smart alert panel shows at least 8 alert types	Mandatory
[ ]	Role switching (Manager / Owner view) is functional	Mandatory
[ ]	Mobile layout tested on 375px width	Mandatory
[ ]	README.md is complete with setup and KPI coverage list	Mandatory
[ ]	Screen recording submitted (3–5 min)	Mandatory
[ ]	Bonus: backend API, real-time updates, export	Optional

7.3 Evaluation Timeline

Stage	Details
Brief Issued	Day 0 — you are reading this document
Clarification Window	Days 1–2 — submit questions via the designated channel
Submission Deadline	Day 7 — end of day (11:59 pm IST)
Review Period	Days 8–10 — evaluator review and scoring

Stage	Details
Demo Presentations	Day 11–12 — shortlisted candidates present live (15 min + 10 min Q&A)
Decision Communicated	Day 14

8. Guidance for Candidates

8.1 What Separates Good from Great

A good submission implements the requirements correctly. A great submission shows that the builder understood WHY those requirements exist — that they thought about the person sitting in front of this dashboard at 8pm on a Friday, trying to figure out why the kitchen is backed up.

The evaluators are hospitality business operators, not just engineers. They will notice if your chart scales make no sense for the data, if your "alert" just shows a number without telling the manager what to do, or if the dashboard looks impressive but answers no real questions.

8.2 Common Mistakes to Avoid

- Building a generic dashboard and relabelling it — the hospitality context must be embedded throughout
- Using randomly generated data that violates real-world patterns (e.g. 100% seat utilisation at 3am)
- Cramming all 49 KPIs onto one screen — hierarchy and navigation are part of the design challenge
- Treating smart alerts as simple threshold indicators with no recommended action copy
- Submitting a localhost-only project — it must be live and publicly accessible
- Ignoring mobile — hospitality managers often check dashboards on their phone between services

8.3 Suggestions for Prioritisation (7-Day Plan)

Day	Focus	Target Output
Day 1	Architecture & Data	Project scaffold, mock data JSON with 90 days of data, routing structure
Day 2	Core KPI Components	KPI card component, domain navigation, time range selector, benchmark logic
Day 3	Revenue & Operations	Revenue domain complete with all charts, operations domain with heatmap
Day 4	Orders, Staff, Customer	Menu engineering quadrant, staff section, customer intelligence section
Day 5	Inventory, ROI, Alerts	Remaining domains + fully functional smart alerts panel
Day 6	Polish & Mobile	Responsive layout fixes, dark mode, role switching, demo alert triggers
Day 7	Deploy & Document	Live deployment, README, screen recording, final submission

8.4 Recommended Reference Resources

- Restaurant industry KPI benchmarks: National Restaurant Association annual report
- Menu engineering methodology: Michael Kasavana & Donald Smith original quadrant framework (1982)
- RevPASH concept: Sheryl Kimes (Cornell School of Hotel Administration)
- Chart library documentation: Recharts.org, ApexCharts.com, Chart.js docs
- Hospitality data context: Toast POS benchmark reports, Lightspeed hospitality insights blog

Good luck. Build something you'd be proud to show a restaurant owner.

Questions? Contact the issuing team through the designated channel before Day 2.